The chain

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Stage 6 of 14 · High concentration · Moderately hard to substitute

Deposition and Etch

Deposition and etch tools build a chip layer by layer. China-based toolmakers' share of deposition tools rose from 2 to 7 percent between 2019 and 2024, by CSET's count.

Specimen 06 · Deposition and Etch

A plasma etch chamber cut in half. The vessel is 500 mm across inside and holds one 300 mm wafer clamped to a holder, called a chuck, at the bottom. The violet haze above the wafer is the plasma. The gap it fills is drawn about twice its real height of 30 to 40 mm. The rest is to scale.

In plain terms

Deposition and etch build the chip itself, one layer at a time. Deposition lays down a thin film, sometimes only a few atoms thick. Lithography prints a pattern on it. Etch then eats away everything the pattern does not protect, and the next film goes down on top. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025[1]. Some of the holes the etch cuts in memory chips are more than 50 times deeper than they are wide, like a finger-wide shaft about a meter deep[2]. The film has to coat them right to the bottom. The etch has to eat one material and leave the one beside it untouched. Applied Materials and Lam Research in the United States, Tokyo Electron in Japan and ASM International in the Netherlands make most of these machines. Chinese toolmakers have gained share in both kinds of tool since 2019.

In short

American companies held 60.3 percent of the world market for deposition tools in 2025, Japanese companies 14.0 percent and China-based companies 9.8 percent, by CSET's count from TechInsights data[3]. In etch and cleaning tools that year, American companies held 52.6 percent, Japanese companies 32.6 percent and China-based companies 10.1 percent[4].

1 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 17. govinfo.gov

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

Atomic layer deposition tools coat a hole's walls and floor as evenly as its rim[5]. China-based companies held only 2.6 percent of the world market for them in 2025, by CSET's count from TechInsights data[6]. The December 2024 US export rule targets tools for atomic layer deposition of the metals molybdenum and ruthenium, and for coating holes more than 200 times deeper than they are wide[7].

Applied Materials and Lam Research are the two largest deposition and etch suppliers. Applied Materials' revenue across all its businesses was $28.368 billion in fiscal 2025[8]. Lam Research's was $23.233 billion in fiscal 2026, the year to June 2026[9].

Chokepoint card — Deposition and Etch

Concentration	Substitutability	Price or market size
High	**Moderate**	**$26.2 billion**
Four to six firms could supply this stage today, and the top two take most of the revenue.	In 2025 Lam Research held 41.1 percent of dry etch tools, Tokyo Electron 23.4 percent and Applied Materials 17.4 percent, by CSET's count from TechInsights data[10].	deposition tool market in 2025, by CSET's count from TechInsights data[3]. Dry etch tools were a $21.9 billion market in 2025[10]

Who leads	US	**Applied Materials**	82.4 percent of physical vapor deposition tools in 2025, by CSET's count from TechInsights data[11]
	US	**Lam Research**	51.2 percent of conductor etch tools in 2025[12]
	JP	**Tokyo Electron**	55.9 percent of dielectric etch tools in 2025[13]
	NL	**ASM International**	54.5 percent of atomic layer deposition tools in 2025[6]
	CN	**Naura**	14.7 percent of physical vapor deposition tools in 2025[11]
Where it is made	US	**United States**	Applied Materials, Lam Research
	JP	**Japan**	Tokyo Electron, Kokusai Electric
	NL	**Netherlands**	ASM International, atomic layer deposition
	KR	**South Korea**	Large equipment buyer
	CN	**China**	Naura, AMEC and Piotech, in the largest equipment market ($49.3 billion in 2025)
Why substitution is possible	Buyers have alternatives[5]. At least five firms sell atomic layer deposition tools, which lay down a film a fraction of a layer of atoms at a time[6].		

3 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 **Export Controls on Semiconductor Manufacturing Items.** U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

6 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

7 **Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for...** U.S. Government Publishing Office, 5 December 2024, p. 33. govinfo.gov

8 **Applied Materials, revenue reported to the SEC (XBRL company data).** U.S. Securities and Exchange Commission, accessed 9 September 2026. data.sec.gov

9 **Lam Research, current report on Form 8-K dated 29 July 2026.** U.S. Securities and Exchange Commission, 29 July 2026. sec.gov

10 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

12 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

13 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

Where China stands

Chinese toolmakers held 12.4 percent of dry etch in 2025, according to CSET's analysis of TechInsights data [10]. They held 2.6 percent of atomic layer deposition that year, by the same count [6].

Where the US stands

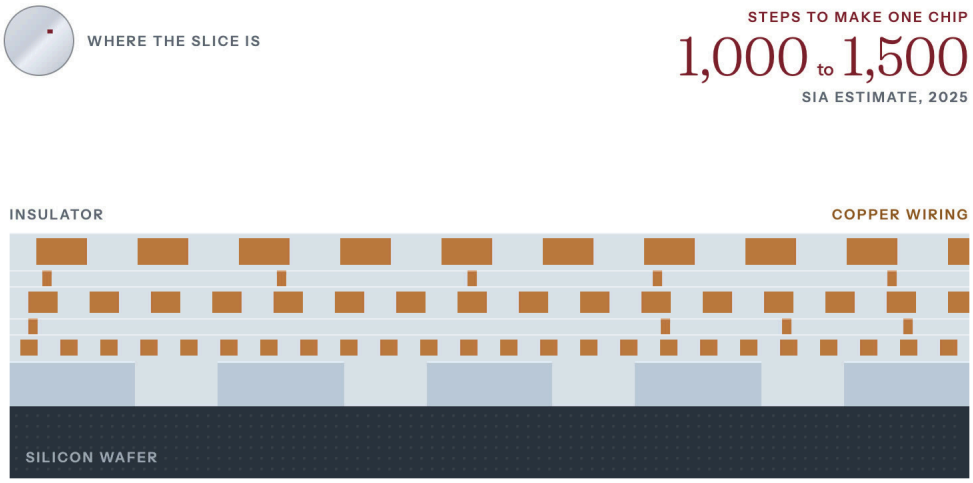
A December 2024 US rule lists some dry etch, atomic layer deposition and deep-hole deposition tools as controlled exports [14]. Its foreign direct product rule reaches some tools built outside the United States [15].

01

How deposition and etching work

Figure 06.1 · Explainer, 24 seconds

One round of deposition and etch



1. **The wafer.** A slice through one spot on a wafer, magnified until its layers show.
2. **Deposit.** A thin film is laid over the whole wafer.
3. **Coat.** A light-sensitive coating, the resist, is spun on top.
4. **Expose.** Light shines through a stencil and changes the resist wherever it lands.
5. **Develop.** A wash removes the resist the light reached, leaving the stencil's pattern.
6. **Etch.** A plasma cuts straight down through the film wherever the resist is gone.
7. **Strip.** The leftover resist is removed. The pattern now sits in the film.
8. **Repeat.** Insulator fills the gaps and the next layer goes on. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Note: Simplified cross-section. The layers are drawn much thicker than they are.

Most etch steps, and many deposition steps, happen in a vacuum chamber that holds one wafer at a time. A pump empties the chamber of air, and a measured flow of gas comes in. A radio-frequency field, an electric field that reverses millions of times a second, turns the gas into a plasma. The field splits the gas into charged ions and into neutral fragments called radicals, which are more reactive than whole molecules [16].

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

16 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com

In most deposition steps, the gas carries the material the film will be made of. The molecules, or the radicals a plasma splits from them, land on the wafer and stick. The film builds up a few atoms at a time.

To coat the floor of a deep hole as thickly as its rim, the machine can work in cycles, a method called atomic layer deposition. First one gas settles in a single layer on every surface, and the pump removes the rest. Then a second gas reacts with that layer and nothing else, and the pump empties the chamber again. Each cycle adds a fraction of a layer of atoms wherever the gas reached, so the film grows as evenly on a hole's walls and floor as on its rim[5].

Not every step runs one wafer at a time. Batch furnaces such as Tokyo Electron's TELINDY PLUS heat as many as 125 wafers at once, usually without plasma[17]. They grow oxide films and run low-pressure chemical vapor deposition and some atomic layer deposition[17].

For etch, the gas attacks one material and leaves the one beside it. Wherever the photoresist covers the wafer, it shields the film beneath. Wherever it does not, the radicals react with the surface and turn it into a gas, which the pump carries away. A second field pulls the ions straight down at the wafer, so the etch cuts vertically and a narrow hole keeps its width as it deepens[16]. When the etch reaches the material underneath, which the gas was chosen not to touch, it slows almost to a stop, and the tool ends the step.

02

Types of deposition and etching

Deposition families

The market sizes and leading shares below are 2025 figures from Georgetown University's Center for Security and Emerging Technology (CSET), based on data from the research firm TechInsights. CSET puts the deposition tool market at $26.2 billion[3]. It puts the dry etch market at $21.9 billion[10].

- **Physical vapor deposition** (PVD), which knocks atoms off a solid target and lands them on the wafer, was a $4.9 billion market in 2025[11]. Applied Materials held 82.4 percent of it and China's Naura 14.7 percent[11].
- **Chemical vapor deposition** (CVD), which reacts gases at the wafer surface to grow a film, was an $11.2 billion market in 2025, led by Applied Materials at 47 percent[18].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

16 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com

17 TELINDY™ Series. Tokyo Electron, accessed 9 September 2026. tel.com

18 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

The plasma-enhanced version (PECVD) runs cooler and made up most of it, $7.5 billion[19].

- **Atomic layer deposition** (ALD) lays down less than one atomic layer at a time. Each step stops once every surface is covered, so it coats vertical walls as evenly as horizontal floors. The US Bureau of Industry and Security (BIS), which writes the export rules, says that even coating makes it important for building 3D structures, such as 3D DRAM, 3D NAND, and the FinFET and gate-all-around transistors in logic chips[5]. ASM International leads it with 54.5 percent of a $4.1 billion market in 2025[6].
- **Epitaxy** grows layers that line up with the crystal beneath them, using high-temperature CVD tools[20]. **Electroplating** deposits the copper used for wiring across the wafer[21].

Etch families

- **Conductor etch**, which cuts gates, metals and silicon, was a $12.2 billion market in 2025, led by Lam at 51.2 percent, by CSET's count from TechInsights data[12].
- **Dielectric etch**, which cuts the insulating oxides and nitrides between the conductors, was an $8.7 billion market in 2025, led by Tokyo Electron at 55.9 percent, by CSET's count from TechInsights data[13].
- **High-aspect-ratio etch** cuts the deep, narrow channels that let 3D NAND stack layers of memory cells[22]. A channel's aspect ratio is its depth divided by its width. Lam says slight departures from a channel's intended shape can hurt yield[22]. Lam runs this etch well below 0 degrees Celsius[22]. It says its Cryo 3.0 technology can etch channels up to 10 microns deep whose width varies from top to bottom by less than 0.1 percent of their depth[22].

AI hardware needs sideways cuts, deeper holes and a second stack of metal wiring on the back of the wafer.

- **Gate-all-around** transistors need highly selective isotropic etch, which cuts sideways[23]. BIS controls isotropic dry etch tools that remove silicon-germanium at least 100 times faster than silicon, a selectivity of 100:1 or more[24].
- **HBM** (high-bandwidth memory) needs holes etched straight through the silicon, called through-silicon vias (TSVs), so chips can be stacked and wired[25]. BIS controls the tools that etch these holes at aspect ratios (depth to width) of 10:1 or more and at

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov
6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
12 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
13 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
19 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
20 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
21 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 39. cset.georgetown.edu
22 Lam Research Introduces Lam Cryo™ 3.0 Cryogenic Etch Technology to Accelerate... Lam Research, 31 July 2024. newsroom.lamresearch.com
23 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 15. govinfo.gov
24 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 28. govinfo.gov
25 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 15. govinfo.gov

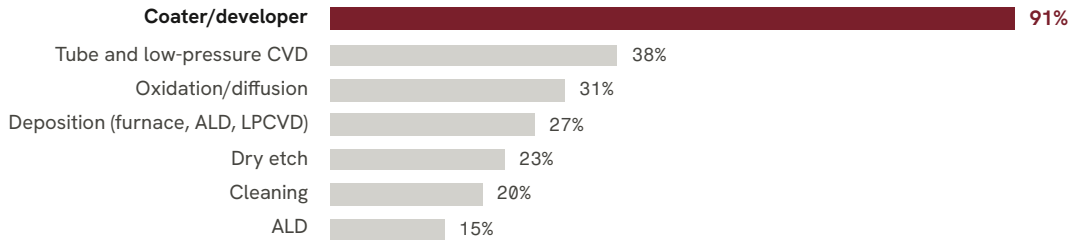
rates above 7 microns a minute[14]. The control covers only tools that also keep the holes' depth even to within 2 percent across the wafer[14].

- **Backside power delivery** moves the wires that carry power to the back of the wafer[26]. In July 2024 Applied Materials said it expected backside power to add $1 billion to the market for its wiring tools, for every 100,000 wafers a month of new fab capacity[27].

Figure 06.2

Tokyo Electron held 91 percent of coater/developers in 2025, on Gartner's figures

Tokyo Electron world market share by tool type, 2025. Some types overlap, because deposition includes furnace, ALD and low-pressure CVD tools



Source: Gartner, as reported in Tokyo Electron's FY2026 Q4 results presentation [28]

03

Who makes deposition and etch tools

The industry association SEMI put world sales of chipmaking equipment at $135.1 billion in 2025, up 15 percent from $117.1 billion[29]. Sales of wafer processing equipment, the category that includes deposition and etch tools, rose 12 percent[29].

No one firm dominates deposition and etch the way ASML dominates lithography. In dry etch, CSET's count for 2025 from TechInsights data puts Lam Research at 41.1 percent, Tokyo Electron at 23.4 percent and Applied Materials at 17.4 percent[10]. Tokyo Electron's own results presentation, citing Gartner, gives it 23 percent of dry etch and 27 percent of deposition systems in 2025, a category that includes furnace, ALD and low-pressure CVD tools[28]. Within that category, Tokyo Electron's share of ALD was 15 percent[28]. The same presentation, again citing Gartner, gives it 91 percent of coater/developers in 2025[28]. These machines coat wafers with photoresist and develop the pattern after lithography.

ASM International, the atomic layer deposition specialist, reached a record EUR 3.2 billion of revenue in 2025 as customers built 2 nm gate-all-around capacity[30]. Molybdenum ALD, a new use of its tools that deposits the metal molybdenum, entered volume production that year[30].

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov

26 PowerVia Test Shows Industry-Leading Performance. Intel, 5 June 2023. download.intel.com

27 Applied Materials Unveils Chip Wiring Innovations for More Energy-Efficient... GlobeNewswire (release by Applied Materials), 8 July 2024. globenewswire.com

28 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 28. tel.com

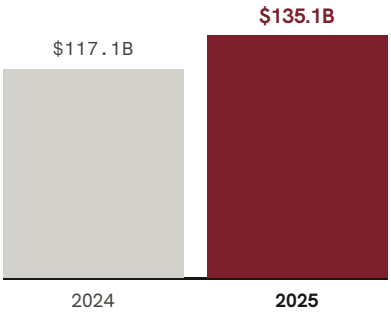
29 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org

30 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com

Figure 06.3

Chipmaking equipment sales rose to $135.1 billion in 2025, from $117.1 billion

What the world spent on chipmaking equipment



Source: SEMI, Worldwide Semiconductor Equipment Market Statistics [29]

04

China's toolmakers and export controls

China is the tool vendors' largest market, buying $49.3 billion of equipment in 2025, half a percent less than in 2024[29]. Taiwan's spending rose 90 percent to $31.5 billion[29]. South Korea's rose 26 percent to $25.8 billion[29]. China still takes a large share of every major vendor's sales.

- **Tokyo Electron**: 34.1 percent of sales in the year to March 2026, down from 41.7 percent[31]. It was 30.4 percent in the quarter to June 2026[32].
- **Lam Research**: about 34 percent of revenue in the year to June 2026, the same as a year earlier[33].
- **Applied Materials**: 28 percent in the third fiscal quarter of 2026, down from 35 percent a year earlier[34].
- **ASM International**: more than 30 percent of 2025 revenue[30].

Between 2019 and 2024, by CSET's analysis of TechInsights data, China-based suppliers went from 2 to 7 percent of deposition and from under 3 to almost 9 percent of etch and clean[35]. In ALD they gained no share and held less than 1 percent in 2024[35].

CSET's newer analysis, of data that TechInsights published in July 2026, puts China-based suppliers at 9.8 percent of deposition in 2025[3]. They held 10.1 percent of etch and

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

29 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org

30 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com

31 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 27. tel.com

32 Q1 FY2027 (April-June 2026) Financial Announcement, transcript. Tokyo Electron, 30 July 2026, p. 6. tel.com

33 Lam Research, annual report on Form 10-K for the fiscal year ended 28 June 2026. U.S. Securities and Exchange Commission, 7 August 2026. sec.gov

34 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials), 13 August 2026. globenewswire.com

35 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

clean that year[4]. In dry etch alone their share was 12.4 percent[10]. In ALD they held 2.6 percent in 2025, most of it from Piotech[6].

The Chinese toolmaker AMEC alone held 12.4 percent of dielectric etch in 2025, by CSET's count from TechInsights data[13]. Its revenue was RMB 12.385 billion that year, up about 36.62 percent[36]. It spent RMB 3.744 billion on research and development, 30.2 percent of sales[36].

The United States tightened its export controls at least five times between 2022 and 2025. The rules describe each deposition or etch tool by the process it is designed to run[24].

- **October 2022.** BIS issued a rule on 7 October that required a license to export advanced chipmaking equipment to China[37]. The rule listed that equipment under a new entry, ECCN 3B090[37]. It also required US persons to get a license to ship, transfer or service items used at advanced fabs in China[38]. It defined these as fabs that make logic chips with non-planar transistors or at 16/14 nm or below, NAND memory with 128 or more layers, or DRAM at 18 nm half-pitch or below[38]. The rule extended US license rules to more foreign-made items bound for 28 Chinese entities on the Entity List[39]. That list names the foreign firms that exporters need a license to supply.
- **October 2023.** A rule issued in October 2023 and effective 17 November rewrote the US control list's entries for chipmaking equipment[40]. It added tools for four processes[5]. The first is dry etch, both sideways (isotropic) and vertical (anisotropic), under ECCN 3B001.c.1[23]. The second is spatial ALD (3B001.d.9), whose design keeps stray particles down to the level advanced chips need[5]. The third is low-fluorine tungsten ALD and CVD (3B001.d.10), which fills deep, narrow gaps in advanced chips with tungsten while keeping resistance low[5]. The fourth is carbon hard mask PECVD (3B001.d.5), which lays down the etch-resistant carbon films that smaller features need[5].
- **December 2024.** BIS controlled 24 more types of equipment and three types of software, and added 140 entities to the Entity List[15]. The new equipment entries

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

13 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

23 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 15. govinfo.gov

24 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 28. govinfo.gov

36 AMEC holds its 2025 annual results briefing (中微公司成功举办2025年度业绩说明会). Advanced Micro-Fabrication Equipment (AMEC), 1 April 2026. amec-inc.com

37 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 7. govinfo.gov

38 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 14. govinfo.gov

39 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 1. govinfo.gov

40 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 1. govinfo.gov

include TSV etch, and deposition into features with aspect ratios above 200:1[41]. BIS also wrote a foreign direct product rule that reaches some tools built outside the United States[15]. The rule largely exempts foreign-made tools exported by firms located in Japan, the Netherlands and the 31 other countries it lists, whose governments can control these tools themselves[42]. Japan, one of those countries, had added advanced chipmaking equipment to its own control list on 23 July 2023[43].

- **September 2025.** BIS removed the Chinese units of Samsung and SK hynix, and Intel Semiconductor (Dalian), from the Validated End-User program, effective 31 December 2025[44]. From that date, US tools bound for them need a license[44]. BIS estimated that the change would add 1,000 license applications a year[44].
- **September 2025.** The Affiliates Rule extended Entity List restrictions to any company at least 50 percent owned by a listed entity[45]. BIS then suspended it until 9 November 2026[46].

BIS enforces the rules on toolmakers too. In February 2026 BIS said Applied Materials had shipped ion implanters, tools that fire atoms into silicon, to a company in China on the Entity List in 2021 and 2022 without the license the Entity List requires[47]. The tools, worth about $126 million, went to China through Applied's Korean unit[47]. BIS set the penalty at about $252 million, twice their value and the most the law allows[47]. In fiscal 2026 Applied Materials took a $253 million charge to settle an export controls compliance matter with BIS[34].

Among deposition and etch tool segments, China-based suppliers have their largest share in dry stripping, which clears leftover photoresist off the wafer[48]. Dry stripping was a $314.6 million market in 2025, by CSET's count from TechInsights data[48]. Japan's Kokusai held 35.7 percent of it, Mattson Technology 22.6 percent and Naura 8.3 percent[48]. Mattson is based in the United States, but Beijing E-Town bought it in 2016[35]. CSET counts Mattson's share as China's[35].

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

34 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials), 13 August 2026. globenewswire.com

35 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

41 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 16. govinfo.gov

42 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 28. govinfo.gov

43 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 21. govinfo.gov

44 Revocation of Validated End-User Authorizations in the People's Republic of... U.S. Government Publishing Office, 2 September 2025. govinfo.gov

45 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. U.S. Government Publishing Office, 30 September 2025, p. 1. govinfo.gov

46 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... U.S. Government Publishing Office, 12 November 2025. govinfo.gov

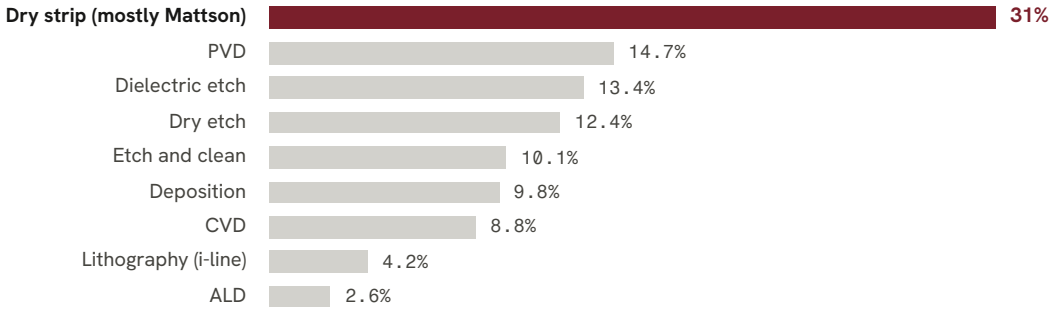
47 Applied Materials to Pay $252 Million Penalty to BIS for Illegally Exporting... U.S. Bureau of Industry and Security, 11 February 2026. bis.gov

48 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

Figure 06.4

China-based suppliers held 31 percent of dry strip in 2025, mostly Mattson's

China-based suppliers' share of global tool segments, 2025



Source: CSET analysis of TechInsights data (published July 2026), ETO Chip Explorer [3]

05

Why deposition and etch are a weaker chokepoint

Deposition and etch tools are a chokepoint, but a weaker one than lithography, because chipmakers can buy each main tool from more than one maker. In dry etch, Lam Research held 41.1 percent in 2025, Tokyo Electron 23.4 percent and Applied Materials 17.4 percent, by CSET's count[10]. Atomic layer deposition has at least five makers[6].

Four questions to answer from memory

1. What do deposition and etch do? see How it works
2. What makes deposition and etch hard? see How it works
3. Who makes most of these tools? see How it works
4. Can a chipmaker switch to another toolmaker here? see Who makes it

Four things to remember

- Deposition lays down thin films and etch cuts them into patterns. Making one chip takes 1,000 to 1,500 steps, by a 2025 industry estimate.
- Four American, Japanese and Dutch firms make most of these tools, and none dominates.
- Chinese makers held 12.4 percent of dry etch and 9.8 percent of deposition in 2025, on CSET's count.
- Chinese makers held 2.6 percent of atomic layer deposition in 2025, most of it from Piotech.

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

Sources (48)

1. A Testimony of David Isaacs, Semiconductor Industry Association, hearing on “Trade in Critical Supply Chains”. U.S. Senate Committee on Finance · 14 May 2025 · p. 2
   www.finance.senate.gov/download/051425-isaacs-testimony&download=1#page=2
2. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 17
   www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=17
3. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
   chipexplorer.eto.tech/?parentNode=N35&selectedNode=N36
4. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
   chipexplorer.eto.tech/?parentNode=N46&selectedNode=N55
5. A Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 · p. 16
   www.govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf#page=16
6. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
   chipexplorer.eto.tech/?parentNode=N35&selectedNode=N41
7. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 33
   www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=33
8. A Applied Materials, revenue reported to the SEC (XBRL company data). U.S. Securities and Exchange Commission (XBRL data for Applied Materials) · Accessed 9 September 2026
   data.sec.gov/api/xbrl/companyconcept/CIK0000006951/us-gaap/RevenueFromContractWithCustomerExcludingAssessedTax.json
9. A Lam Research, current report on Form 8-K dated 29 July 2026. U.S. Securities and Exchange Commission (filing by Lam Research) · 29 July 2026
   www.sec.gov/Archives/edgar/data/707549/000070754926000033/lrcx_exhibitx991xq4x2026.htm
10. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N46&selectedNode=N103
11. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N35&selectedNode=N42
12. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N46&selectedNode=N50
13. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N46&selectedNode=N51
14. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 32
    www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=32
15. A Commerce Strengthens Export Controls to Restrict China's Capability to Produce Advanced Semiconductors for Military Applications. U.S. Bureau of Industry and Security · 2 December 2024
    www.bis.gov/press-release/commerce-strengthens-export-controls-restrict-chinas-capability-produce-advanced-semiconductors-military
16. A Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research · 12 June 2024
    newsroom.lamresearch.com/etch-essentials-semiconductor-manufacturing
17. A TELINDY™ Series. Tokyo Electron · Accessed 9 September 2026
    www.tel.com/product/telindy.html
18. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N35&selectedNode=N47
19. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N35&selectedNode=N38
20. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
    chipexplorer.eto.tech/?parentNode=N35&selectedNode=N40
21. A The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 · p. 39
    cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf#page=39

22. A Lam Research Introduces Lam Cryo™ 3.0 Cryogenic Etch Technology to Accelerate Scaling of 3D NAND for the AI Era. Lam Research · 31 July 2024
newsroom.lamresearch.com/2024-07-31-Lam-Research-Introduces-Lam-Cryo-TM-3-0-Cryogenic-Etch-Technology-to-Accelerate-Scaling-of-3D-NAND-for-the-AI-Era

23. A Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 · p. 15
www.govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf#page=15

24. A Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 · p. 28
www.govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf#page=28

25. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 15
www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=15

26. A PowerVia Test Shows Industry-Leading Performance. Intel · 5 June 2023
download.intel.com/newsroom/archive/2025/en-us-2023-06-05-powervia-test-shows-industryleading-performance.pdf

27. A Applied Materials Unveils Chip Wiring Innovations for More Energy-Efficient Computing. GlobeNewswire (release by Applied Materials) · 8 July 2024
www.globenewswire.com/news-release/2024/07/08/2909540/0/en/applied-materials-unveils-chip-wiring-innovations-for-more-energy-efficient-computing.html

28. A FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron · 30 April 2026 · p. 28
www.tel.com/ir/library/report/pjuomj00000000tf-att/fy26q4transcript-e.pdf#page=28

29. A SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in 2025, Up 15% Year-on-Year. SEMI · 7 April 2026
www.semi.org/en/SEMI-Reports-Global-Semiconductor-Equipment-Billings-Reached-135-Billion-in-2025

30. A ASM reports Q4 and full-year 2025 results. ASM International · 3 March 2026 · p. 2
www.asm.com/media/yvxbavwe/20260303-asm-reports-q4-and-full-year-2025-results.pdf#page=2

31. A FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron · 30 April 2026 · p. 27
www.tel.com/ir/library/report/pjuomj00000000tf-att/fy26q4transcript-e.pdf#page=27

32. A Q1 FY2027 (April-June 2026) Financial Announcement, transcript. Tokyo Electron · 30 July 2026 · p. 6
www.tel.com/ir/library/report/i242su0000000goz-att/fy27q1transcript-e.pdf#page=6

33. A Lam Research, annual report on Form 10-K for the fiscal year ended 28 June 2026. U.S. Securities and Exchange Commission (filing by Lam Research) · 7 August 2026
www.sec.gov/Archives/edgar/data/707549/000070754926000037/lrcx-20260628.htm

34. A Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials) · 13 August 2026
www.globenewswire.com/news-release/2026/08/13/3344890/0/en/applied-materials-announces-third-quarter-2026-results.html

35. A Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025
cset.georgetown.edu/article/inside-beijings-chipmaking-offensive/

36. A AMEC holds its 2025 annual results briefing (中微公司成功举办2025年度业绩说明会). Advanced Micro-Fabrication Equipment (AMEC) · 1 April 2026
www.amec-inc.com/news/708.html

37. A Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification. U.S. Government Publishing Office (Federal Register) · 13 October 2022 · p. 7
www.govinfo.gov/content/pkg/FR-2022-10-13/pdf/2022-21658.pdf#page=7

38. A Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification. U.S. Government Publishing Office (Federal Register) · 13 October 2022 · p. 14
www.govinfo.gov/content/pkg/FR-2022-10-13/pdf/2022-21658.pdf#page=14

39. A Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification. U.S. Government Publishing Office (Federal Register) · 13 October 2022 · p. 1
www.govinfo.gov/content/pkg/FR-2022-10-13/pdf/2022-21658.pdf#page=1

40. A Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 · p. 1
www.govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf#page=1

41. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 16
www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=16

42. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 28
www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=28

43. A Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 · p. 21
www.govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf#page=21

44. A **Revocation of Validated End-User Authorizations in the People's Republic of China**. U.S. Government Publishing Office (Federal Register) · 2 September 2025
www.govinfo.gov/content/pkg/FR-2025-09-02/html/2025-16735.htm

45. A **Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities**. U.S. Government Publishing Office (Federal Register) · 30 September 2025 · p. 1
www.govinfo.gov/content/pkg/FR-2025-09-30/pdf/2025-19001.pdf#page=1

46. A **One Year Suspension of Expansion of End-User Controls for Affiliates of Certain Listed Entities**. U.S. Government Publishing Office (Federal Register) · 12 November 2025
www.govinfo.gov/content/pkg/FR-2025-11-12/pdf/2025-19846.pdf

47. A **Applied Materials to Pay $252 Million Penalty to BIS for Illegally Exporting Semiconductor Manufacturing Equipment**. U.S. Bureau of Industry and Security · 11 February 2026
www.bis.gov/press-release/applied-materials-pay-252-million-penalty-bis-illegally-exporting-semiconductor-manufacturing-equipment

48. A **Supply Chain Explorer: Advanced Chips**. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
chipexplorer.eto.tech/?parentNode=N46&selectedNode=N53

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

How to cite this chapter

> Sharma, Yashvardhan. "Deposition and Etch." *Chip Supply Chain: How AI Chips Are Made*. Edition of 29 October 2026. https://chipsupplychain.org/deposition-and-etch/

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@misc{chipsupplychain2026depositionandetch,
  title        = {Deposition and Etch},
  author       = {Sharma, Yashvardhan},
  howpublished = {Chip Supply Chain: How AI Chips Are Made},
  year         = {2026},
  month        = {October},
  note         = {Edition of 29 October 2026},
  url          = {https://chipsupplychain.org/deposition-and-etch/}
}
```